

A Project Report
on
**SOCIRANK: IDENTIFYING AND RANKING PREVALENT
NEWS TOPICS USING SOCIAL MEDIA FACTORS**

Submitted in Partial fulfillment of the requirement for the award of degree
of

Bachelor of Engineering

in

Computer Science and Engineering

by

CH. SOWMYA 1608-15-733-004

N. CHANDINI 1608-15-733-008

S. UDAYASRI 1608-15-733-020

Under the guidance of

Mr. M. PRAVEEN KUMAR

Assistant Professor

Dept. of CSE.



Department of Computer Science and Engineering

Matrusri Engineering College

(Affiliated to Osmania University, Approved by AICTE)

Saidabad, Hyderabad, T.S-500059
(2025-2026)

Department of Computer Science and Engineering

Matrusri Engineering College

(Affiliated to Osmania University, Approved by AICTE)

Saidabad, Hyderabad, T.S-500059

(2025-2026)



CERTIFICATE

This is to certify that the project entitled “**SocioRank: Identifying and Ranking prevalent news topics using social media factors**” submitted by **Ms. Ch. Sowmya, Ms. N. Chandini** and **Ms. S. Udayasri** bearing **HT.No: 1608-15-733-004, 1608-15-733-008 and 1608-15-733-020** in the partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering** is a bonafide work carried by them.

The results of the investigations enclosed in this report have been verified and found satisfactory.

Project Guide

H.O.D.

Mr. M. Praveen Kumar

Assistant Professor

Dept. of CSE

Dr.T.Raghunadha Reddy

Associate Professor

Dept. of CSE

External Examiner(s)

ACKNOWLEDGEMENT

This project consumed huge amount of work, research and dedication. Still implementation would not have been possible if we did not have support from our Project Guide, Project Coordinator's, Head of the Department and Principal. Therefore, we like to extend our sincere gratitude to all of them.

We are grateful to our project guide **Mr. M. Praveen Kumar**, Assistant Professor, Dept. of CSE, Matrusri Engineering College for provision of expertise, technical support and guidance in the implementation.

We wish to express our gratitude to project coordinator **Mrs. M. Swapna Reddy** for their indefatigable inspiration, constructive criticisms and encouragement throughout this dissertation work.

We would like to express our sincere thanks to the Associate Professor and Head of the Department, **Dr. T. Raghunadha Reddy**, for permitting us to do this project.

We would like to express our gratitude to **Dr. D. Hanumantha Rao**, principal of Matrusri Engineering College who permitted to carry out this project as per the academics.

We would like to thank **CSE** Department for providing us this opportunity to share and contribute our part of work to accomplish the project in time and all the teaching and support staff for their steadfast support and encouragement.

Nevertheless, we express our gratitude towards our families and colleagues for their kind cooperation and encouragement which helped us in completion of this project.

DECLARATION

We hereby declare that the project entitled “**SocioRank: Identifying and Ranking prevalent news topics using social media factors**” is submitted to Osmania University in the partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering**.

This is a record of the bonafide work carried out by us under the guidance of **Mr. M. Praveen Kumar**, Assistant Professor, Matrusri Engineering College, Saidabad, Hyderabad. The Results embodied in this report have not been reproduced/copied from any source. The results embodied in this report have not been submitted to any other University or Institute for the award of any other degree or diploma.

Ch. Sowmya 1608-15-733-004

N. Chandini 1608-15-733-008

S. Udayasri 1608-15-733-020

Dept of CSE, MECS.

ABSTRACT

Mass media sources, specifically the news media, have traditionally informed us of daily events. In modern times, social media services such as Twitter provide an enormous amount of user-generated data, which have great potential to contain informative news-related content. **(Derek Davis, Gerardo Figueroa, and Yi-Shin Chen, iee transactions on systems, man, and cybernetics: systems.)** For these resources to be useful, we must find a way to filter noise and only capture the content that, based on its similarity to the news media, is considered valuable. However, even after noise is removed, information overload may still exist in the remaining data—hence, it is convenient to prioritize it for consumption. To achieve prioritization, information must be ranked in order of estimated importance considering three factors. First, the temporal prevalence of a particular topic in the news media is a factor of importance, and can be considered the media focus (MF) of a topic. Second, the temporal prevalence of the topic in social media indicates its user attention (UA). Last, the interaction between the social media users who mention this topic indicates the strength of the community discussing it, and can be regarded as the user interaction (UI) toward the topic. We propose an unsupervised framework—SociRank—which identifies news topics prevalent in both social media and the news media, and then ranks them by relevance using their degrees of MF, UA, and UI. Our experiments show that SociRank improves the quality and variety of automatically identified news topics.

Present System:

Historically, knowledge that appries the general public of daily events has been provided by mass media sources, specifically the news media. The news media presents professionally verified occurrences or events, while social media presents the interests of the audience in these areas, and may thus provide insight into their popularity. Unfortunately,filter noise and only capture the content that, based on its news media, and social media is very difficult. However, even after noise is removed, information overload may still exist in the remaining data. Hence, it is difficult to prioritize.

Disadvantages:

1. Hard to find a way to filter news from noisy.
2. High computational demand to prioritize.

Suggest System:

We propose an unsupervised system SociRank which effectively identifies news topics that are prevalent in both social media and the news media, and then ranks them by relevance using

their degrees of MF, UA, and UI. Even though this paper focuses on news topics. News media sources are considered reliable because they are published by professional journalists, who are held accountable for their content. On the other hand, the Internet, being a free and open forum for information exchange, has recently seen a fascinating phenomenon known as social media. In social media, regular, non-journalist users are able to publish unverified content and express their interest in certain events.

Consolidated, filtered, and ranked news topics from both professional news providers and individuals have several benefits. The most evident use is the potential to improve the quality and coverage of news recommender systems or Web feeds, adding user popularity feedback.

Advantages:

1. We can find a way to filter noise and only capture the news.
2. We can filter the news based on topic.
3. Main use potential to improve the quality and coverage of news recommender systems.

CONTENTS

S. No	Chapter	Page No.
1.	Introduction	1
2.	Literature Survey	3
3.	System Analysis	8
4.	System Design	12
	4.1. Class Diagram	12
	4.2. Usecase Diagram	13
	4.3. Sequence Diagram for User	14
	4.4. Sequence Diagram for Admin	15
	4.5. Collaboration Diagram for User	16
	4.6. Collaboration Diagram for Admin	16
	4.7. State Chart Diagram	17
	4.8. Activity Diagram	18
	4.9. Component Diagram	19
	4.10. Deployment Diagram	20
	4.11. E-R Diagram	21
	4.12. Data Flow Diagram	22
5.	Implementation	28
	5.1. Algorithm	28
	5.2. Software Requirements	29
	5.3. Hardware Requirements	30
	5.4. Source Code	30
6.	System Testing	37
7.	Results	42
8.	Conclusion	50
9.	References	51

LIST OF FIGURES

S. No.	Fig. No	Name of the figure	Page No.
--------	---------	--------------------	----------